INFORMATION SYSTEMS

Assignment A

JANUARY 25, 2019

18024092
Fealy

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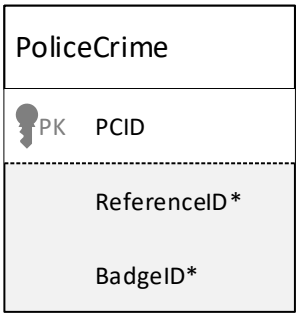
Class Diagram

Classes with datasets

Police

Attribute	Data type
BadgeID	Auto generate number
Title	Short text
First Name	Long text
Middle Initial	Short text
Surname	Long text
Rank	Long text
Address	Long text
Phone number	Number
Email Address	Hyperlink
Date of Birth	Date/time

PoliceCrime



Attribute	Data type
PCID	Auto generate number
ReferenceID*	Number
BadgeID*	Number

GeoLocation

GeoLocation	
PK	LocationID
	Name
	Postcode
	Population
	%Unemployed
	%Employed
	%Students
	%Single parent families
	%Home Owner
	%Car owner

Attribute	Data type
LocationID	Auto generate number
Name	Long text
Postcode	Long text
Population	Number
%Unemployed	Number
%Employed	Number
%Students	Number
%Single Parent Families	Number
%Home Owner	Number
%Car Owner	Number

PoliceLocation

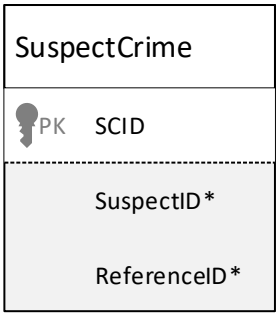
PoliceLocation	
PK	PLID
	BadgeID*
	LocationID*

Attribute	Data type
PLID	Auto generate number
BadgeID*	Number
LocationID*	Number

Suspect

Attribute	Data type
SuspectID	Auto generate number
LocationID*	Number
First Name	Long text
Last Name	Long Text
Date Of Birth	Date/time
Address	Long text
Phone number	Number
Previous Convictions	Long text

SuspectCrime



Attribute	Data type
SCID	Auto generate number
SuspectID*	Number
ReferenceID*	Long text

Victim

Victim	
PK	VictimID
	LocationID*
	First Name
	Last Name
	Date Of Birth
	Phone number
	Email Address

Attribute	Data type
VictimID	Auto generate number
LocationID*	Number
First Name	Long text
Last Name	Long text
Date Of Birth	Date/time
Phone number	Number
Email Address	Hyperlink

VictimCrime

VictimCrime	
PK	VCID
	VictimID*
	ReferenceID*
	Statement

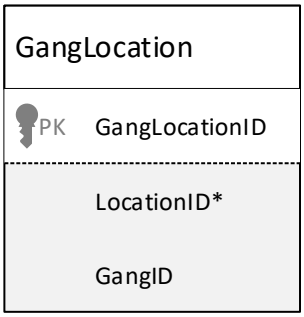
Attribute	Data type
VCID	Auto generate number
VictimID*	Number
ReferenceID*	Number
Statement	Long text

Gang

Gang	
PK	GangID
	SuspectID*
	GangLocationID*
	Gang Name

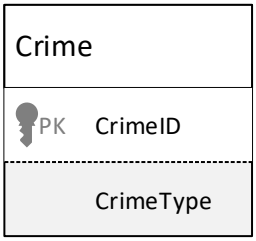
Attribute	Data type
GangID	Auto generate number
SuspectID*	Number
GangLocationID*	Number
Gang Name	Long text

GangLocation



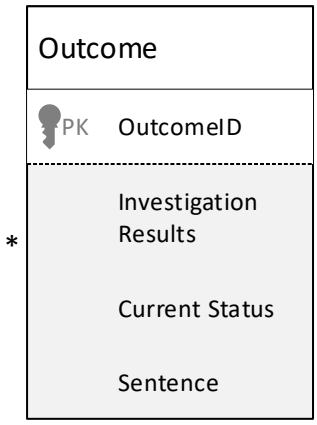
Attribute	Data type
GangLocationID	Auto generate number
LocationID*	Number
GangID*	Number

Crime



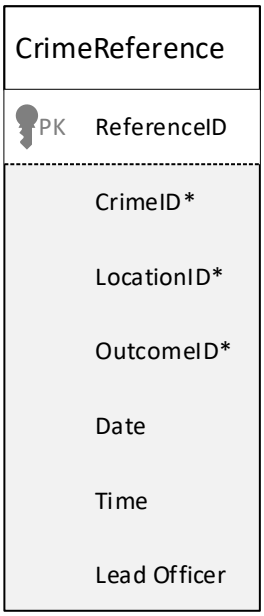
Attribute	Data type
CrimeID	Auto generate number
Crime Type	Long text

Outcome



Attribute	Data type
OutcomeID	Auto generate number
Status	Long Text

CrimeReference



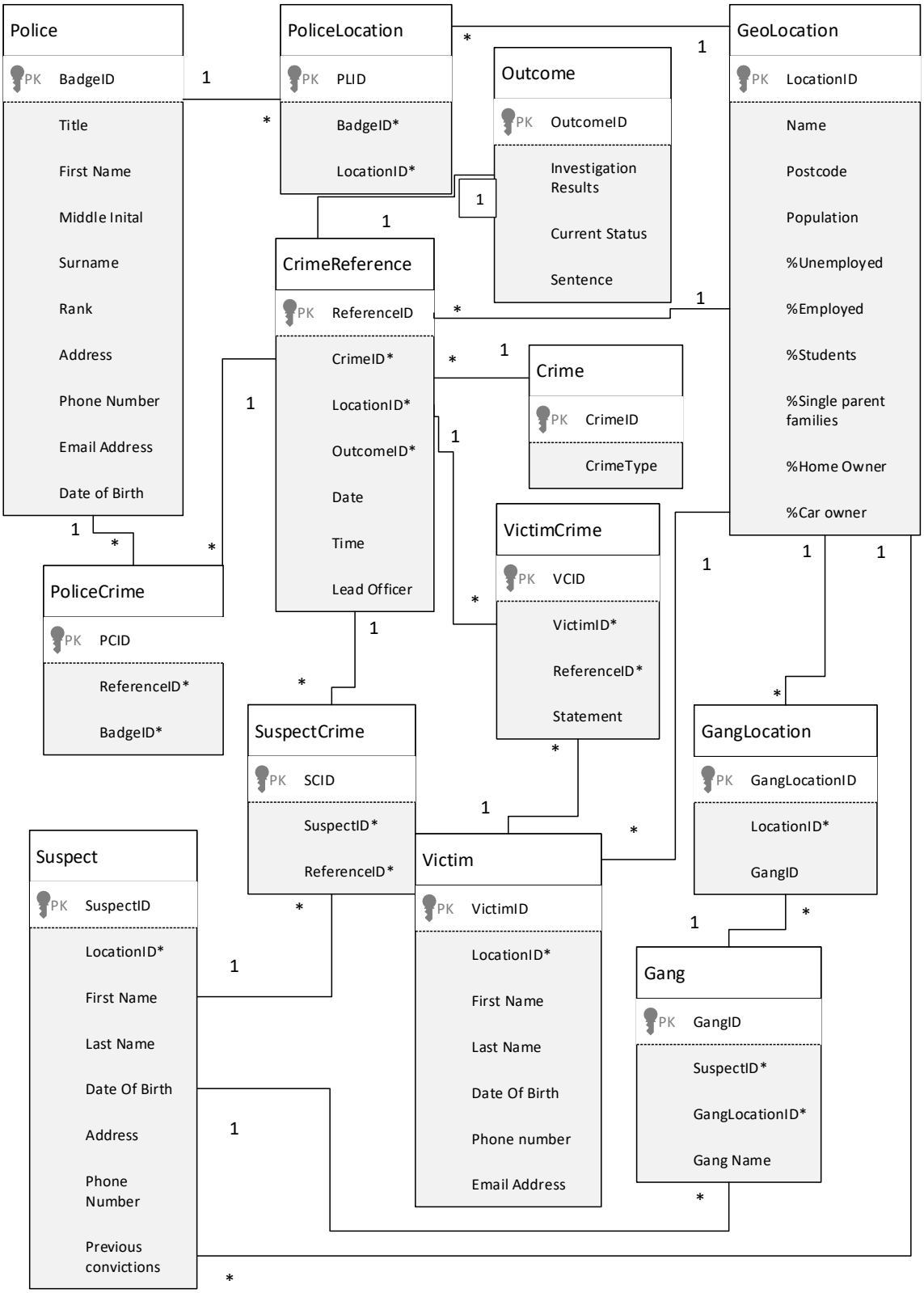
Attribute	Data type
ReferenceID	Auto generate number
CrimeID*	Number
LocationID*	Number
OutcomeID*	Number
Date	Date/time
Time	Date/time
Lead officer	Long text

Relationship assumptions

Class	Relationship	Class	Assumption
Police	1:M	PoliceLocation	1 officer can only be in 1 location, but locations can have many officers
PoliceLocation	M:1	GeoLocation	Each policeloaction can only assigned one GeoLocation, but geolocation can be assigned to many different Policelocations
Police	1:M	PoliceCrime	One Officer can have one PoliceCrime, but PoliceCrime can have many officers
CrimeReference	M:1	Crime	CrimeReference can have many Crimes but Crime can only have one CrimeReference
Suspect	1:M	SuspectCrime	SuspectCrime can only have one Suspect, but Suspect can have many SuspectCrime
CrimeReference	1:1	Outcome	A CrimeReference can only be assigned to one Outcome. And one outcome can only be assigned to one CrimeReference

Victim	1:M	VictimCrime	A victim can be in many VictimCrime, but only one VictimCrime will relate to one Victim
Suspect	1:M	Gang	A suspect can only be in one gang, but gangs can have many suspects
Gang	1:M	GangLocation	A gang can only be in one location, but there a location can have many gangs
GangLocation	M:1	GeoLocation	There can be many GangLocation to a GeoLocation, but there can only be one GeoLocation to a GangLocation
CrimeReference	1:M	PoliceCrime	PoliceCrime can only have CrimeRefence, but CrimeRefernece can have many PoliceCrime
CrimeReference	1:M	SuspectCrime	SuspectCrime can only have one CrimeRefence, but CrimeRefernece can have many SuspectCrime
CrimeReference	1:M	VictimCrime	VictimCrime can only have one CrimeRefence, but CrimeRefernece can have many VictimCrime
CrimeReferecne	M:1	GeoLocaotion	CrimeReference can only have one location, but one GeoLocation can have many CrimeReferences
Suspects	M:1	GeoLocation	A suspect can only be in one Location, but one location can have many suspects
Victim	M:1	GeoLocation	A victim can only be in one location, but location can have many victims

Class Diagram



MS Access

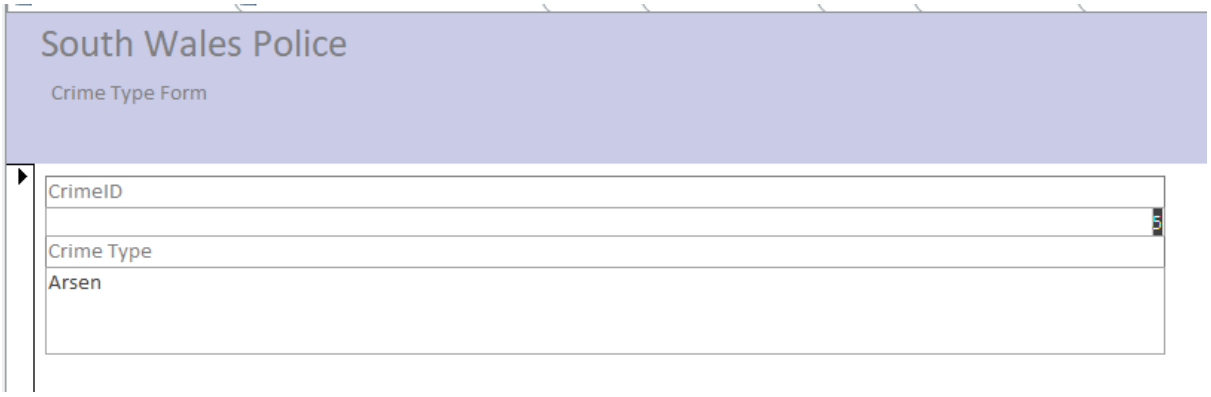
Tables

Crime

Table

CrimeID	Crime Type	Click to Add
1	Murder	
2	Sexual Assault	
3	Rape	
4	Breaking and Entering	
5	Arsen	
6	Theft	
7	Car theft	
8	GBA	
9	Terrorism	
10	Treason	
(New)		

Form



Report

CrimeReport

CrimeID
1
Crime Type
Murder

CrimeID
2
Crime Type
Sexual Assault

CrimeID
3
Crime Type
Rape

CrimeID
4
Crime Type
Breaking and Entering

Crime Reference

Table

ReferenceID	CrimeID*	LocationID*	Date	Time	Lead Officer	Cli
1	1	1	05/01/2019	01:40:29	John Smith	
2	3	2	21/01/2019	06:05:02	John Smith	
3	6	4	30/12/2018	12:05:01	John Smith	
4	6	4	29/12/2018	16:50:20	Jenny Jamesons	
5	6	4	20/01/2019	20:40:06	Jenny Jamesons	
6	5	3	05/04/2018	21:04:02	Chloe Burns	
7	4	2	05/10/2018	22:05:02	David Davids	
8	4	1	06/11/2018	06:40:40	Senior Davids	
9	4	7	09/01/2019	07:04:20	Rob Jones	
10	2	1	17/01/2019	15:47:52	Tracy Lockey	
11	2	8	21/12/2018	16:58:52	Tracy Lockey	
12	6	7	18/12/2018	17:46:10	Senior Davids	
13	4	1	06/01/2001	19:20:01	Senior Davids	
(New)	0	0				

Form

South Wales Police

Crime Reference Form

ReferenceID	CrimeID*	LocationID*	Date	TIme
1	1	1	05/01/2019	01:40:29
Lead Officer				
John Smith				

Report

Crime Reference Report

ReferenceID	CrimeID*	LocationID*	Date	TIme
1	1	1	05/01/2019	01:40:29
Lead Officer				
John Smith				

ReferenceID	CrimeID*	LocationID*	Date	TIme
2	3	2	21/01/2019	06:05:02
Lead Officer				
John Smith				

ReferenceID	CrimeID*	LocationID*	Date	TIme
3	6	4	30/12/2018	12:05:01
Lead Officer				
John Smith				

ReferenceID	CrimeID*	LocationID*	Date	TIme
4	6	4	29/12/2018	16:50:20
Lead Officer				
Jenny Jamesons				

Gang

Table

GangID*	GangLocatic	Gang Name	Click to Add
1	2	Tinter Tots	
2	3	Electric Army	
3	4	Foreign frontier	
4	5	Helios	
5	6	Cardiff Massive	
6	7	Merthyr Classics	
7	1	Pontypridd Ballers	
(New)	0		

Form

South Wales Police

Gang Form

GangID*	GangLocationID*
1	2

Gang Name

Tinter Tots

Report

Gang Report

GangID*	GangLocationID*
1	2

Gang Name

Tinter Tots

GangID*	GangLocationID*
2	3

Gang Name

Electric Army

GangID*	GangLocationID*
3	4

Gang Name

Foreign frontier

GangID*	GangLocationID*
4	5

Gang Name

Helios

GangLocation

Table

GangLocatic	LocationID*	GangID*	Click
5	1	4	
4	3	3	
1	4	7	
3	5	2	
2	6	1	
7	8	6	
6	9	5	
* (New)	0	0	

Form

South Wales Police

Gang Location Form

GangLocationID	LocationID*	GangID*
5	1	4

Report

Gang Location Report

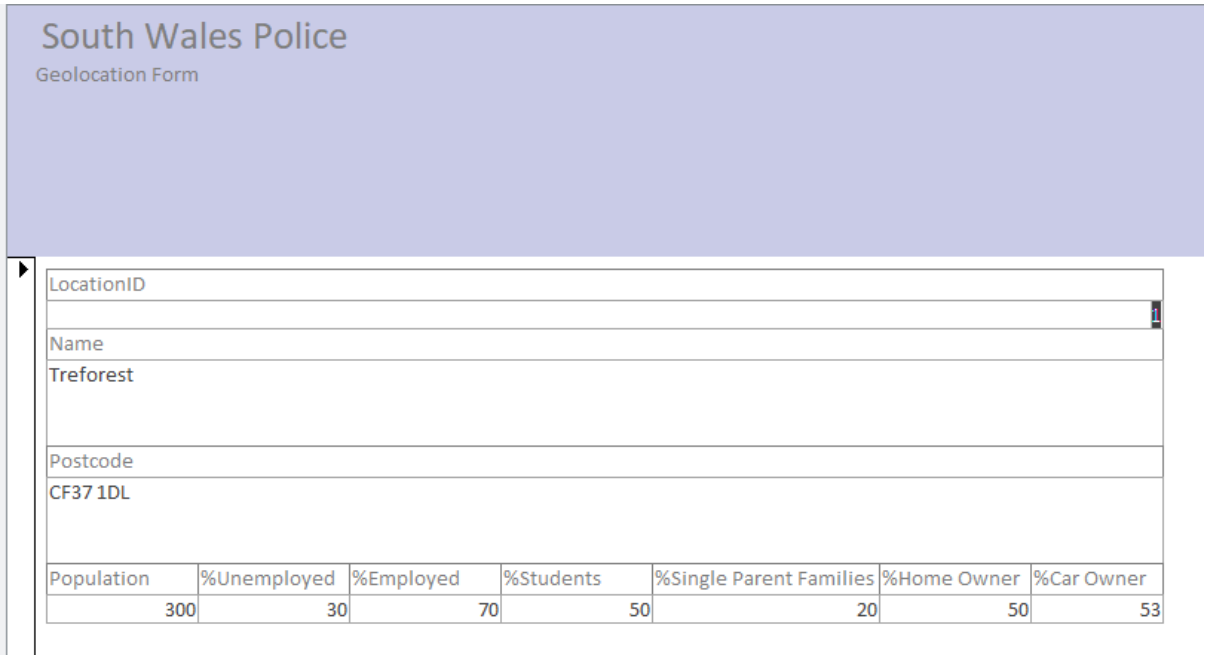
GangLocationID	LocationID*	GangID*
5	1	4
4	3	3
1	4	7
3	5	2
2	6	1
7	8	6
6	9	5

GeoLocation

Table

LocationID	Name	Postcode	Population	%Unemploy	%Employed	%Students	%Single Par	%Home Ow	%Car Owner	Cli
1	Treforest	CF37 1DL	300	30	70	50	20	50	53	
2	Pontypridd	CF37 1HE	100	40	60	30	40	20	40	
3	Treforest	CF37 1EE	200	54	46	35	10	40	30	
4	Pontypridd	CF37 4TG	236	10	90	10	30	62	70	
5	Caerphilly	CF83 1NX	200	23	77	20	15	59	62	
6	Caerphilly	CF83 1HG	423	50	50	35	40	30	61	
7	Merthyr Tydfil	CF47 0AY	300	30	70	26	31	69	85	
8	Merthyr Tydfil	CF48 2AF	500	20	80	31	9	50	60	
9	Cardiff	CF1 1AA	300	20	80	6	30	50	90	
(New)			0	0	0	0	0	0	0	

Form



Report

GeoLocation Report

LocationID						
1						
Name						
Treforest						
Postcode						
CF37 1DL						
Population	%Unemployed	%Employed	%Students	%Single Parent Families	%Home Owner	%Car Owner
300	30	70	50	20	50	53

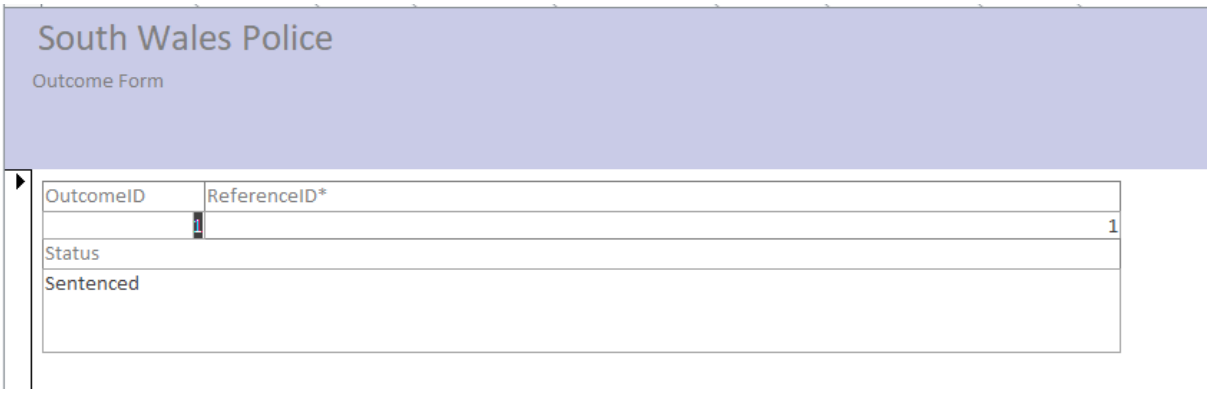
LocationID						
2						
Name						
Pontypridd						
Postcode						
CF37 1HE						
Population	%Unemployed	%Employed	%Students	%Single Parent Families	%Home Owner	%Car Owner
100	40	60	30	40	20	40

Outcome

Table

OutcomeID	ReferenceID	Status	Click to Add
1	1	Sentenced	
2	2	Investigating	
3	3	Investigating	
4	4	Sentenced	
5	5	Closed	
6	6	In court	
7	7	Sentenced	
8	8	Investigating	
9	9	Closed	
10	10	In court	
11	11	Sentenced	
12	12	Sentenced	
13	13	Sentenced	
* (New)	0		

Form


South Wales Police
Outcome Form
OutcomeID
ReferenceID*
1
1
Status
Sentenced

Report

Outcome Report

OutcomeID	ReferenceID*
1	1
Status	
Sentenced	

OutcomeID	ReferenceID*
2	2
Status	
Investigating	

OutcomeID	ReferenceID*
3	3
Status	
Investigating	

OutcomeID	ReferenceID*
4	4
Status	
Sentenced	

Police

Table

BadgeID	Title	First Name	Middle Inita	Surname	Rank	Address	Date Of Birt	Email Addre	Click
1	Mr	John	J	Smith	Sergeant	123 Peppercor	06/06/1987	JJSmith@live.c	
2	Mrs	Jenny	J	Jamesons	Special	52,	26/01/1985	JJJ@live.com	
3	Mrs	Chloe		Burns	Police constab	23	25/10/1991	ChloeBurns@h	
4	Mr	Senior		Davids	Chief inspecto	34	03/04/1956	SeniorDavids@	
5	Mr	Rob	S	Jones	Police constab	18,	06/08/1967	RobJones@liv	
6	Miss	Niamh	J	Bennett	Chief superint	56,	16/04/1983	NiamhBennett	
7	Mr	David	H	Davids	Inspector	7,	31/07/1963	DavidDavids@	
8	Mrs	Tracy	E	Lockey	Police constab	45,	06/09/1978	TracyLockey@	
(New)									

Form

South Wales Police

Police Form

BadgeID	1
Title	Mr
First Name	John
Middle Inital	J
Surname	Smith
Rank	Sergeant
Address	123 Peppercorn Lane
Date Of Birth	06/06/1987
Email Address	JJSmith@live.com

Report

Police Report

BadgeID	1
Title	Mr
First Name	John
Middle Inital	J
Surname	Smith
Rank	Sergeant
Address	123 Peppercorn Lane
Date Of Birth	06/06/1987
Email Address	JJSmith@live.com

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PoliceCrime

Table

PCLID	ReferenceID*	BadgeID*	Click to Add
1	1	1	
2	2	1	
3	3	1	
4	4	2	
5	5	2	
6	6	3	
7	7	7	
8	8	4	
9	9	5	
10	10	8	
11	11	8	
12	12	4	
14	13	4	
(New)	0	0	

Form

South Wales Police

Police Crime Form

PCLID	ReferenceID*	BadgeID*
1	1	1

Report

Suspect Crime Report

SCID	SuspectID*	ReferenceID*
1	1	1
2	2	1
3	3	2
4	4	3
5	5	4
6	6	5
7	7	6
8	8	7
9	9	8
10	10	9
11	11	10
12	11	11

PoliceLocation

Table

PLID	BadgeID*	LocationID*	Click to Add
1	1	1	
2	2	3	
3	3	5	
4	4	8	
5	5	9	
6	6	7	
7	7	2	
8	8	4	
* (New)	0	0	

Form

South Wales Police

Police Location Form

PLID	BadgeID*	LocationID*
1	1	1

Report

Police Location Report

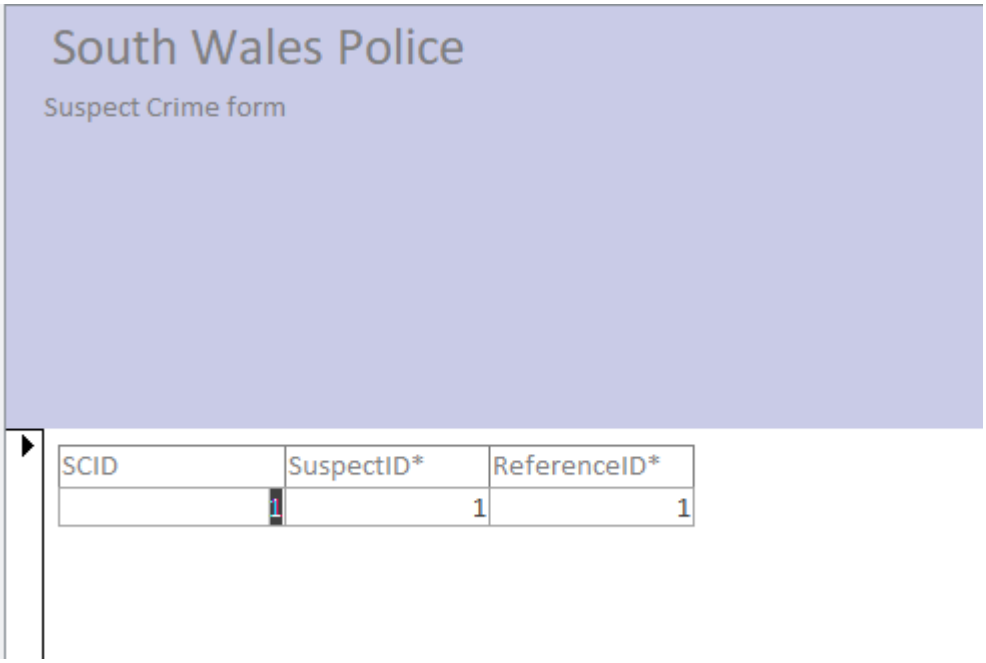
PLID	BadgeID*	LocationID*
1	1	1
2	2	3
3	3	5
4	4	8
5	5	9
6	6	7
7	7	2
8	8	4

SuspectCrime

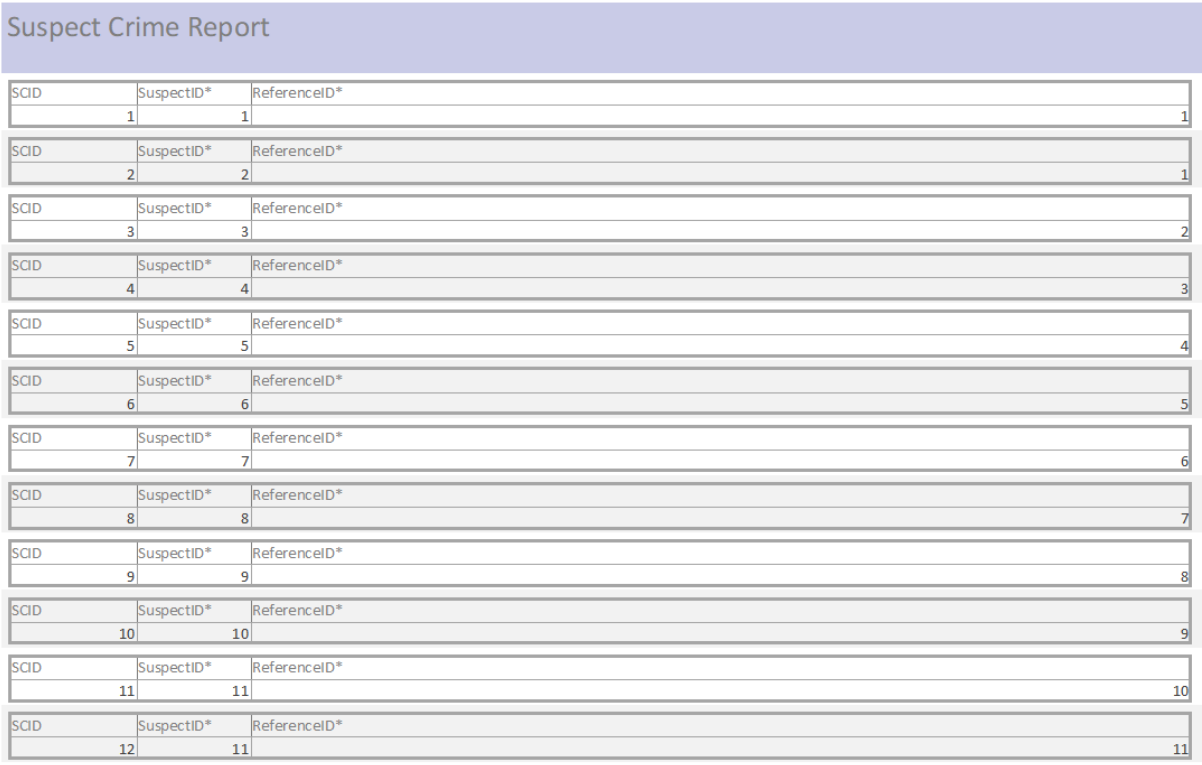
Table

SCID	SuspectID*	ReferenceID	Click to Add
1	1	1	
2	2	1	
3	3	2	
4	4	3	
5	5	4	
6	6	5	
7	7	6	
8	8	7	
9	9	8	
10	10	9	
11	11	10	
12	11	11	
(New)	0	0	

Form



Report



Suspect Crime Report

SCID	SuspectID*	ReferenceID*
1	1	1
2	2	1
3	3	2
4	4	3
5	5	4
6	6	5
7	7	6
8	8	7
9	9	8
10	10	9
11	11	10
12	11	11

Victim

Table

VictimID	LocationID*	First Name	Last Name	Date of Birtl	Phone Num	Email Address	Click to Add
1	2	Chloe	Jones	09/05/1997	36265652	Chloe@live.com	
2	1	Jane	Smith	05/02/1978	16546516	JS@live.com	
3	5	John	Smith	06/07/1963	15456431	JohnSmith@live.com	
4	4	James	Jameson	19/11/1959	65464656	JamesJameson@outlook.com	
5	3	Isabella	Ince	08/10/1999	54654321	IsabellaInce@outlook.com	
6	7	Jack	Whitehall	09/10/1987	46545646	JackWhitehall@AOL.com	
7	6	Rhianna	Cole	06/12/1986	44132430	RhiCole@yahoo.com	
8	9	Sam	Smith	08/07/1996	10546984	SamSmith@outlook.com	
9	8	Harry	Kent	14/10/2005	54646876	HarryKent@live.com	
10	3	Pen	Low	31/05/1989	65465402	PenLow@aol.com	
11	4	Flow	High	25/01/1969	55456456	FlowHigh@live.com	
(New)	0				0		

Form

South Wales Police

Victim Form

VictimID	LocationID*
1	2

First Name

Chloe

Last Name

Jones

Date of Birth	Phone Number
09/05/1997	36265652

Email Address

Chloe@live.com

Report

Victim Report

VictimID	LocationID*
1	2

First Name

Chloe

Last Name

Jones

Date of Birth	Phone Number
09/05/1997	36265652

Email Address

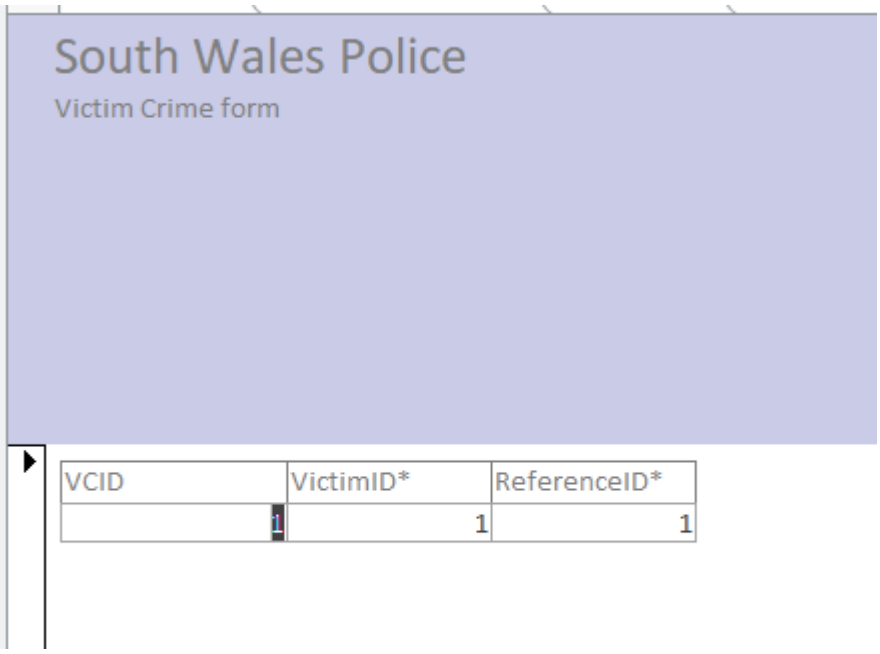
Chloe@live.com

VictimCrime

Table

VCID	VictimID*	ReferenceID	Click to Add
1	1	1	
2	2	2	
3	3	3	
4	4	4	
5	5	5	
6	6	6	
7	7	7	
8	8	8	
9	10	10	
10	10	10	
11	11	11	
* (New)	0	0	

Form

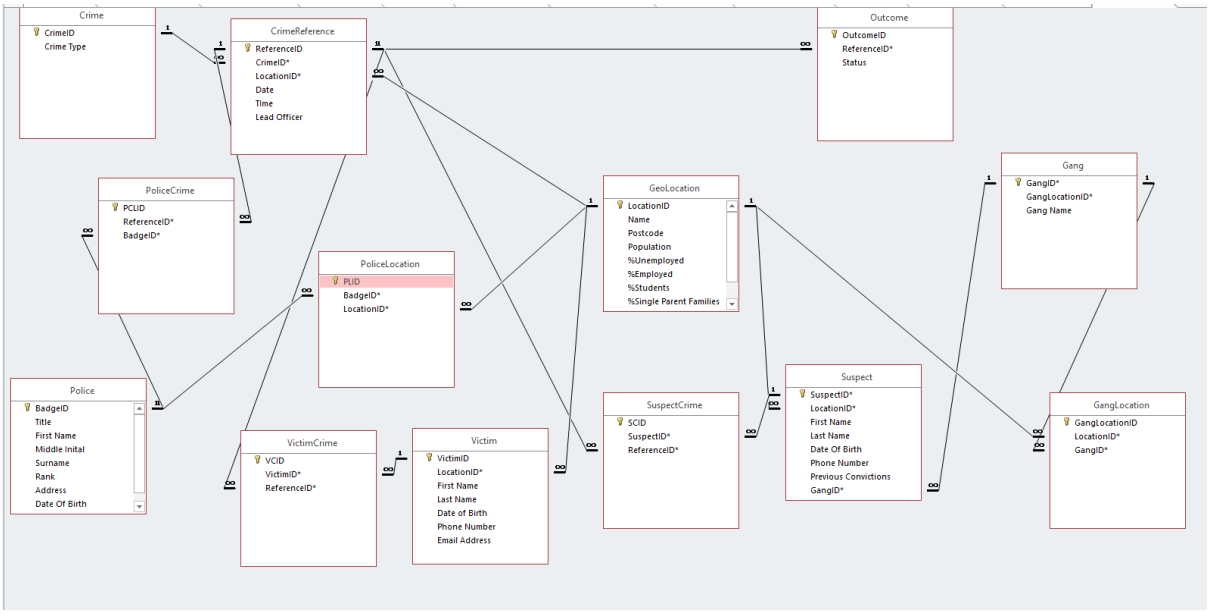


Report

Victim Crime Report

VCID	VictimID*	ReferenceID*
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	10	10
10	10	10
11	11	11

Relationships

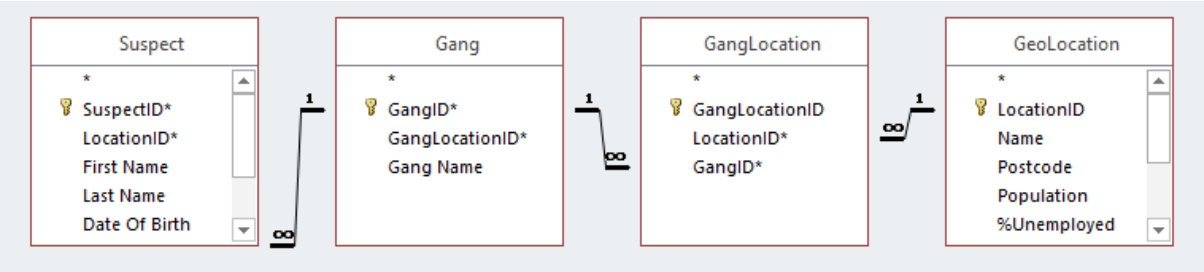


Queries

Query one – Suspects In Gang

The main use of this query is to use the relations created between Suspect and Gang to create a list where all suspects are placed next to their corresponding gang in the geolocation. The output is a report with 5 attributes, suspect ID, suspect first name, suspect last name, gang ID, gang name, and location name. This data come from the corresponding field from suspect, gang, gang location, and geo location.

Tables



Fields

Field:	SuspectID*	First Name	Last Name	GangID*	Gang Name	Name
Table:	Suspect	Suspect	Suspect	Gang	Gang	GeoLocation
Sort:						
Show:	☑	☑	☑	☑	☑	☑
Criteria:						
or:						

SQL

SELECT Suspect.[SuspectID*], Suspect.[First Name], Suspect.[Last Name], Gang.[GangID*], Gang.[Gang Name], GeoLocation.Name

FROM GeoLocation INNER JOIN ((Gang INNER JOIN Suspect ON Gang.[GangID*] = Suspect.[GangID*]) INNER JOIN GangLocation ON Gang.[GangID*] = GangLocation.[GangID*]) ON GeoLocation.LocationID = GangLocation.[LocationID*];

Report

South Wales Police

Query: Assoicates Police officers with their current cases

ReferenceID*	BadgeID	First Name	Surname	Rank	Crime Type	Status
1	1	John	Smith	Sergeant	Murder	Sentenced
2	1	John	Smith	Sergeant	Rape	Investigating
3	1	John	Smith	Sergeant	Theft	Investigating
4	2	Jenny	Jamesons	Special	Theft	Sentenced
5	2	Jenny	Jamesons	Special	Theft	Closed
6	3	Chloe	Burns	Police constable	Arsen	In court
7	7	David	Davids	Inspector	Breaking and Entering	Sentenced
8	4	Senior	Davids	Chief inspector	Breaking and Entering	Investigating
9	5	Rob	Jones	Police constable	Breaking and Entering	Closed
10	8	Tracy	Lockey	Police constable	Sexual Assault	In court
11	8	Tracy	Lockey	Police constable	Sexual Assault	Sentenced
12	4	Senior	Davids	Chief inspector	Theft	Sentenced
13	4	Senior	Davids	Chief inspector	Breaking and Entering	Sentenced

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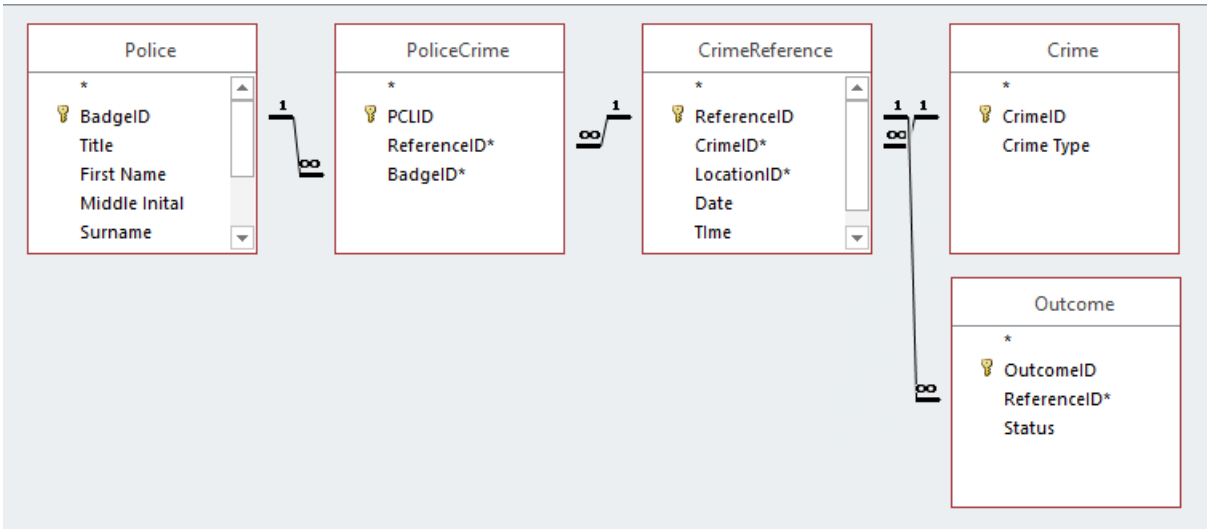
Query Two – Police Officers with their cases

This Query is used to show all data from Police offices and the cases they are the lead officer on and also displays the outcomes linked to those cases. This query utilises 5 different tables to generate a report with 7 attributes. These attributes are BadgeID, First Name, Surname, Rank, ReferenceID, crime type, and status. These attributes from their respective fields from, Police, CrimeReference, Crime, and outcome.

Tables



Fields

Field:	BadgeID	First Name	Surname	Rank	ReferenceID*	Crime Type	Status
Table:	Police	Police	Police	Police	CrimeReference	Crime	Outcome
Sort:							
Show:	☑	☑	☑	☑	☑	☑	☑

SQL

SELECT Police.BadgeID, Police.[First Name], Police.Surname, Police.Rank, CrimeReference.[ReferenceID*], Crime.[Crime Type], Outcome.Status

FROM (Crime INNER JOIN (CrimeReference INNER JOIN (Police INNER JOIN PoliceCrime ON Police.BadgeID = PoliceCrime.[BadgeID*]) ON CrimeReference.ReferenceID = PoliceCrime.[ReferenceID*]) ON Crime.CrimeID = CrimeReference.[CrimeID*]) INNER JOIN Outcome ON CrimeReference.ReferenceID = Outcome.[ReferenceID*];

Report

South Wales Police

Query: Associates suspects with their crimes

SuspectID*	First Name	Last Name	Date Of Birth	ReferenceID	Lead Officer	Crime Type
1	Bob	Foxtrot	08/10/1965	1	John Smith	Murder
2	Adam	Smith	09/02/1963	1	John Smith	Murder
3	John	Adams	06/05/1989	2	John Smith	Rape
4	Jenny	Smith	05/04/1978	3	John Smith	Theft
5	Jessie	Coal	07/08/1963	4	Jenny Jamesons	Theft
6	Bessie	Jones	16/10/1953	5	Jenny Jamesons	Theft
7	James	Pinder	16/10/1998	6	Chloe Burns	Arsen
8	Jamie	Olive	31/05/1997	7	David Davids	Breaking and Entering
9	Brenda	Bennett	06/08/1991	8	Senior Davids	Breaking and Entering
10	Olivia	Cole	09/04/1999	9	Rob Jones	Breaking and Entering
11	Jasmine	Bent	07/08/1985	10	Tracy Lockey	Sexual Assault
11	Jasmine	Bent	07/08/1985	11	Tracy Lockey	Sexual Assault

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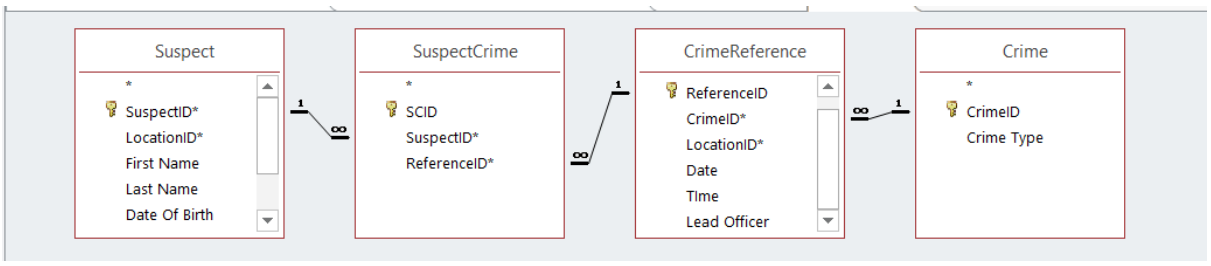
Query Three – Suspects with their crimes

This query is used to associate suspects with their crimes committed. This query utilises 3 main classes, and one joiner class. From these classes 7 attributes are used, SuspectID, First Name, Last Name, Date Of Birth, ReferenceID, Lead officer, and Crime Type.

Tables



Fields

Field:	SuspectID*	First Name	Last Name	Date Of Birth	ReferenceID	Lead Officer	Crime Type
Table:	Suspect	Suspect	Suspect	Suspect	CrimeReference	CrimeReference	Crime
Sort:							
Show:	☑	☑	☑	☑	☑	☑	☑
Criteria:							
or:							

SQL

SELECT Suspect.[SuspectID*], Suspect.[First Name], Suspect.[Last Name], Suspect.[Date Of Birth], CrimeReference.ReferenceID, CrimeReference.[Lead Officer], Crime.[Crime Type]

FROM Crime INNER JOIN (CrimeReference INNER JOIN (Suspect INNER JOIN SuspectCrime ON Suspect.[SuspectID*] = SuspectCrime.[SuspectID*]) ON CrimeReference.ReferenceID = SuspectCrime.[ReferenceID*]) ON Crime.CrimeID = CrimeReference.[CrimeID*];

Report

South Wales Police

Query: Assosiates suspect with their gang

SuspectID*	First Name	Last Name	GangID*	Gang Name	Location Name
11	Jasmine	Bent	6	Merthyr Classics	Merthyr Tydfil
4	Jenny	Smith	6	Merthyr Classics	Merthyr Tydfil
9	Brenda	Bennett	4	Helios	Treforest
6	Bessie	Jones	4	Helios	Treforest
2	Adam	Smith	4	Helios	Treforest
1	Bob	Foxtrot	4	Helios	Treforest
3	John	Adams	3	Foreign frontier	Treforest
10	Olivia	Cole	2	Electric Army	Caerphilly
5	Jessie	Coal	2	Electric Army	Caerphilly
7	James	Pinder	1	Tinter Tots	Caerphilly
8	Jamie	Olive	7	Pontypridd Ballers	Pontypridd

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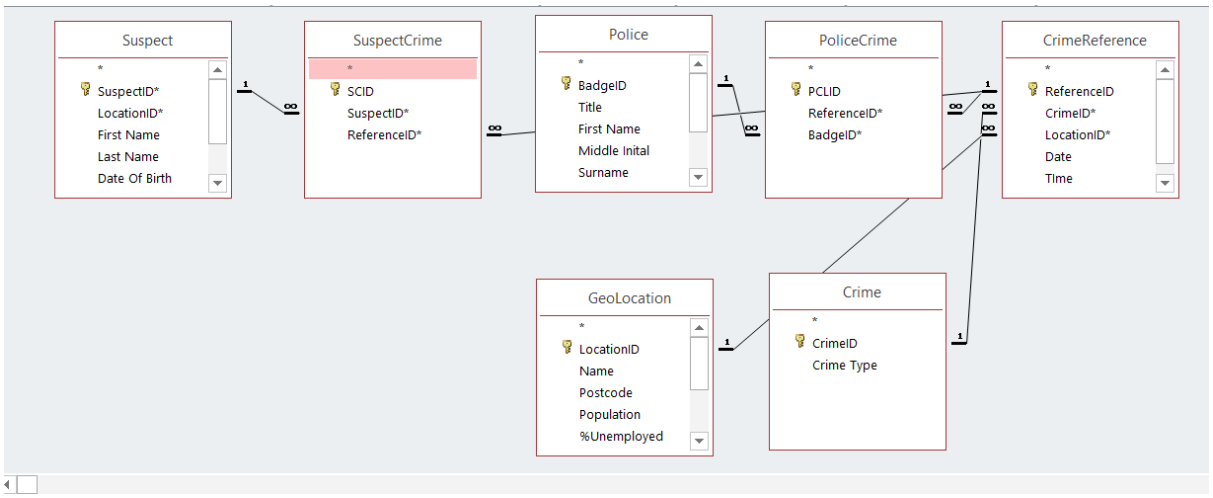
Query 4 – Crimes That Occurred In 2018

This query lists all reported crimes that occurred in 2018, listed with the suspect, location, crime type, assigned officers, and date. This query used 7 Classes, 2 of those classes are only joiner classes. This query also utilises 10 attributes from these classes. To easily sort which crimes were commited in 2018, the criteria for the Date (This information comes from Crime Reference) has >#01/01/2018# and <31/12/2018# this allows for the sorting for data.

Tables



Fields

Field:	SuspectID*	First Name	Last Name	ReferenceID	Date	Crime Type	Name	BadgeID	First Name	Surname
Table:	Suspect	Suspect	Suspect	CrimeReference	CrimeReference	Crime	GeoLocation	Police	Police	Police
Sort:										
Show:	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑
Criteria:					>#01/01/2018# And <#31/12/2018#					
or:										

SQL

SELECT Suspect.[SuspectID*], Suspect.[First Name], Suspect.[Last Name], CrimeReference.ReferenceID, CrimeReference.Date, Crime.[Crime Type], GeoLocation.Name, Police.BadgeID, Police.[First Name], Police.Surname

FROM GeoLocation INNER JOIN (Crime INNER JOIN ((CrimeReference INNER JOIN (Police INNER JOIN PoliceCrime ON Police.BadgeID = PoliceCrime.[BadgeID*]) ON CrimeReference.ReferenceID = PoliceCrime.[ReferenceID*]) INNER JOIN (Suspect INNER JOIN SuspectCrime ON Suspect.[SuspectID*] = SuspectCrime.[SuspectID*]) ON CrimeReference.ReferenceID = SuspectCrime.[ReferenceID*]) ON Crime.CrimeID = CrimeReference.[CrimeID*]) ON GeoLocation.LocationID = CrimeReference.[LocationID*]

WHERE (((CrimeReference.Date)>#1/1/2018# And (CrimeReference.Date)<#12/31/2018#));

Report

South Wales Police

Query: Crimes in 2018

SuspectID*	First Name	Last Name	ReferenceID	Date	Crime Type	Location Name	BadgeID	First Name	Surname
11	Jasmine	Bent	11	21/12/2018	Sexual Assault	Merthyr Tydfil	8	Tracy	Lockey
9	Brenda	Bennett	8	06/11/2018	Breaking and Entering	Treforest	4	Senior	Davids
8	Jamie	Olive	7	05/10/2018	Breaking and Entering	Pontypridd	7	David	Davids
7	James	Pinder	6	05/04/2018	Arsen	Treforest	3	Chloe	Burns
5	Jessie	Coal	4	29/12/2018	Theft	Pontypridd	2	Jenny	Jamesons
4	Jenny	Smith	3	30/12/2018	Theft	Pontypridd	1	John	Smith

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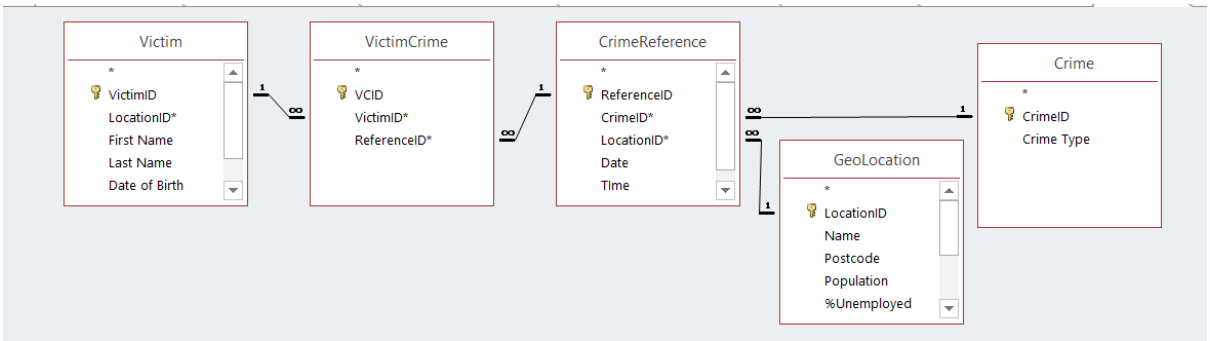
Query 5 – Shows all victims with their reported crimes

This query is used to show all victim with their reported crimes, this is used to easily identify which victim has had what crimes committed against them. This query uses

Tables



Fields

Field:	VictimID	First Name	Last Name	ReferenceID	Name	Date	TIme	Crime Type
Table:	Victim	Victim	Victim	CrimeReference	GeoLocation	CrimeReference	CrimeReference	Crime
Sort:								
Show:	☑	☑	☑	☑	☑	☑	☑	☑
Criteria:								

SQL

SELECT Victim.VictimID, Victim.[First Name], Victim.[Last Name], CrimeReference.ReferenceID, GeoLocation.Name, CrimeReference.Date, CrimeReference.TIme, Crime.[Crime Type]

FROM GeoLocation INNER JOIN (Crime INNER JOIN (CrimeReference INNER JOIN (Victim INNER JOIN VictimCrime ON Victim.VictimID = VictimCrime.[VictimID*]) ON CrimeReference.ReferenceID = VictimCrime.[ReferenceID*]) ON Crime.CrimeID = CrimeReference.[CrimeID*]) ON GeoLocation.LocationID = CrimeReference.[LocationID*];

Report

South Wales Police

Query: Victims with associated crimes

VictimID	First Name	Last Name	ReferenceID	Name	Date	TIme	Crime Type
1	Chloe	Jones	1	Treforest	05/01/2019	01:40:29	Murder
2	Jane	Smith	2	Pontypridd	21/01/2019	06:05:02	Rape
3	John	Smith	3	Pontypridd	30/12/2018	12:05:01	Theft
4	James	Jameson	4	Pontypridd	29/12/2018	16:50:20	Theft
5	Isabella	Ince	5	Pontypridd	20/01/2019	20:40:06	Theft
6	Jack	Whitehall	6	Treforest	05/04/2018	21:04:02	Arsen
7	Rhianna	Cole	7	Pontypridd	05/10/2018	22:05:02	Breaking and Entering
8	Sam	Smith	8	Treforest	06/11/2018	06:40:40	Breaking and Entering
10	Pen	Low	10	Treforest	17/01/2019	15:47:52	Sexual Assault
10	Pen	Low	10	Treforest	17/01/2019	15:47:52	Sexual Assault
11	Flow	High	11	Merthyr Tydfil	21/12/2018	16:58:52	Sexual Assault

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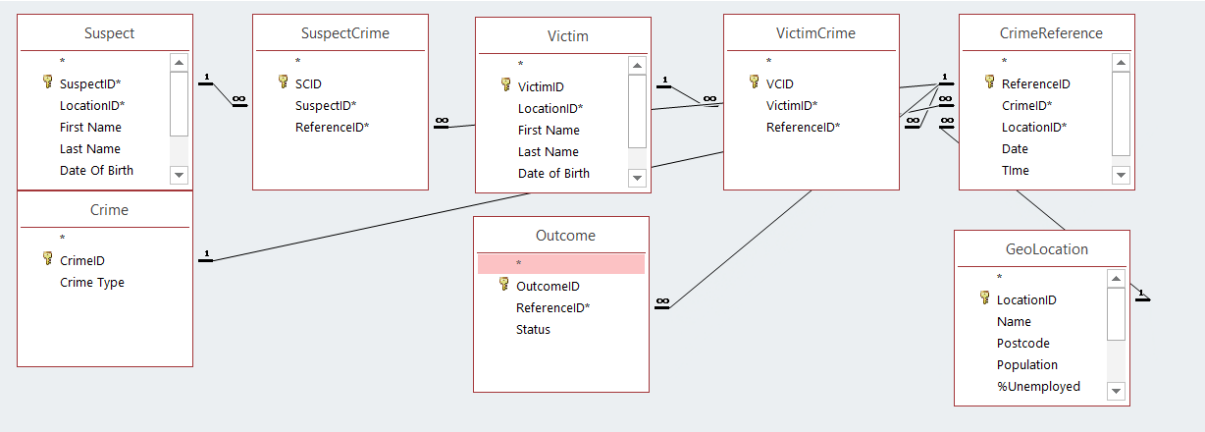
Query 6 – All theft crimes that have been reported

This query is used to identify all theft that have been reported. This query used 8 different classes to achieve this. And takes 10 attributes from these classes respectively. To ensure only theft crimes are displayed, under the crime type criteria ‘ Theft ‘ has been entered that means only theft crimes will be displayed.

Tables



Fields

Field:	SuspectID*	First Name	Last Name	VictimID	First Name	Last Name	ReferenceID	Crime Type	Name	Status
Table:	Suspect	Suspect	Suspect	Victim	Victim	Victim	CrimeReference	Crime	GeoLocation	Outcome
Sort:										
Show:	☑	☑	☑	☑	☑	☑	☑	☑	☑	☑
riteria:										
or:								'Theft'		

SQL

SELECT Suspect.[SuspectID*], Suspect.[First Name], Suspect.[Last Name], Victim.VictimID, Victim.[First Name], Victim.[Last Name], CrimeReference.ReferenceID, Crime.[Crime Type], GeoLocation.Name, Outcome.Status

FROM (GeoLocation INNER JOIN (Crime INNER JOIN ((CrimeReference INNER JOIN (Suspect INNER JOIN SuspectCrime ON Suspect.[SuspectID*] = SuspectCrime.[SuspectID*]) ON CrimeReference.ReferenceID = SuspectCrime.[ReferenceID*]) INNER JOIN (Victim INNER JOIN VictimCrime ON Victim.VictimID = VictimCrime.[VictimID*]) ON CrimeReference.ReferenceID = VictimCrime.[ReferenceID*]) ON Crime.CrimeID = CrimeReference.[CrimeID*]) ON GeoLocation.LocationID = CrimeReference.[LocationID*]) INNER JOIN Outcome ON CrimeReference.ReferenceID = Outcome.[ReferenceID*]

WHERE (((Crime.[Crime Type])='Theft'));

Report

South Wales Police

Query: All Theft Crimes

SuspectID*	First Name	Last Name	VictimID	First Name	Last Name	ReferenceID	Crime Type	Name	Status
4	Jenny	Smith	3	John	Smith	3	Theft	Pontypridd	Investigating
5	Jessie	Coal	4	James	Jameson	4	Theft	Pontypridd	Sentenced
6	Bessie	Jones	5	Isabella	Ince	5	Theft	Pontypridd	Closed

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